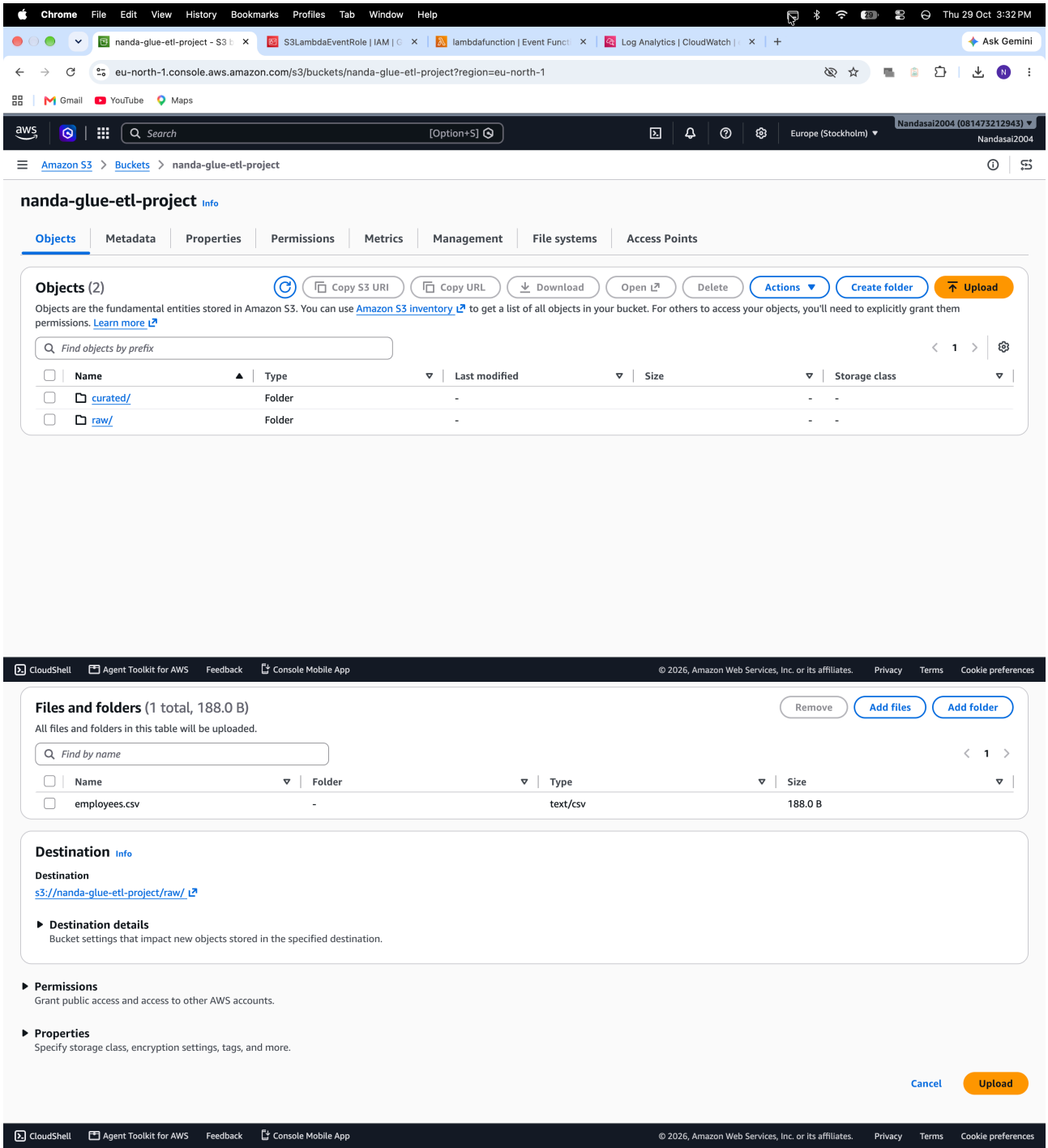
AWS Glue ETL Pipeline with Amazon S3

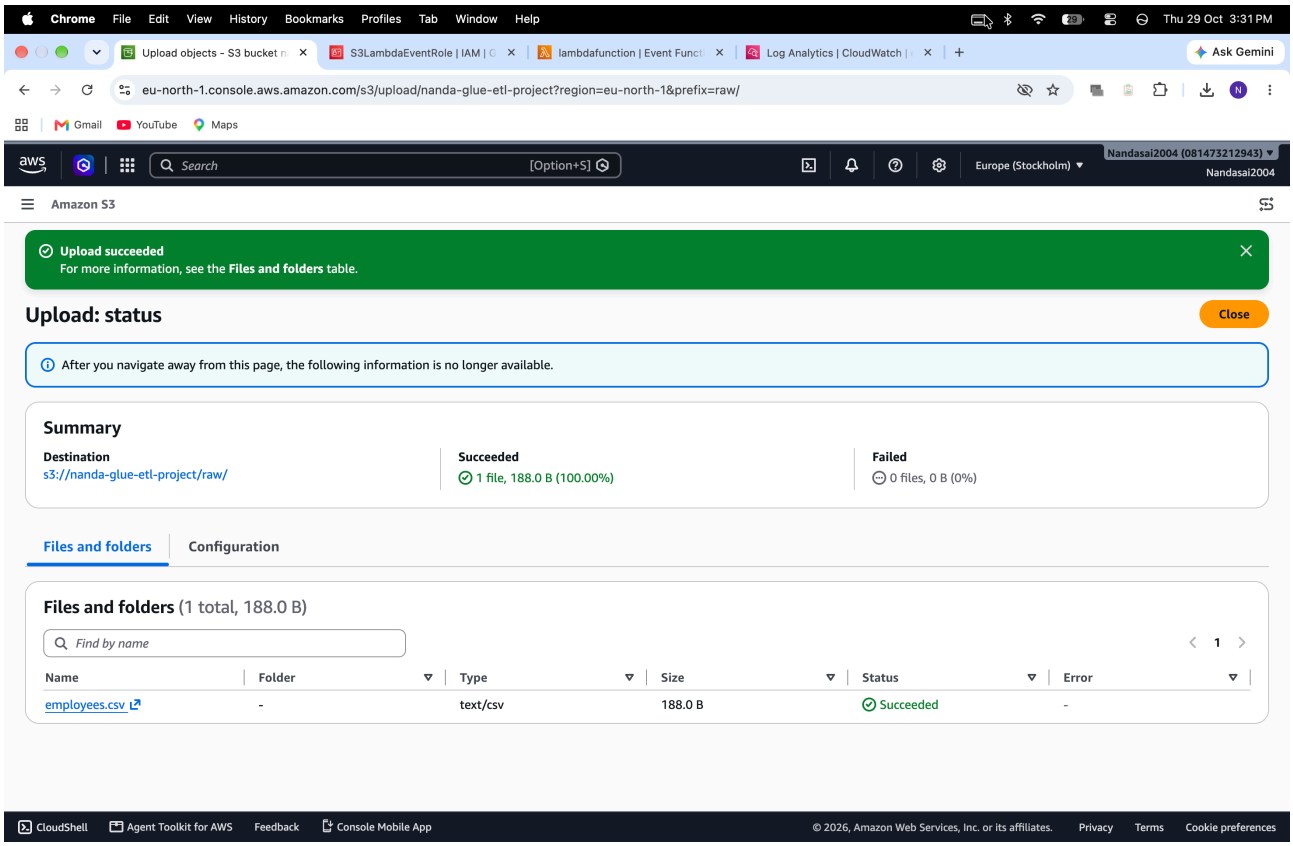
NAME : GORLAMANDALA NANDA SAI
USN : 23BTRCL042
HCLID : HCLTFP2262836

Aim

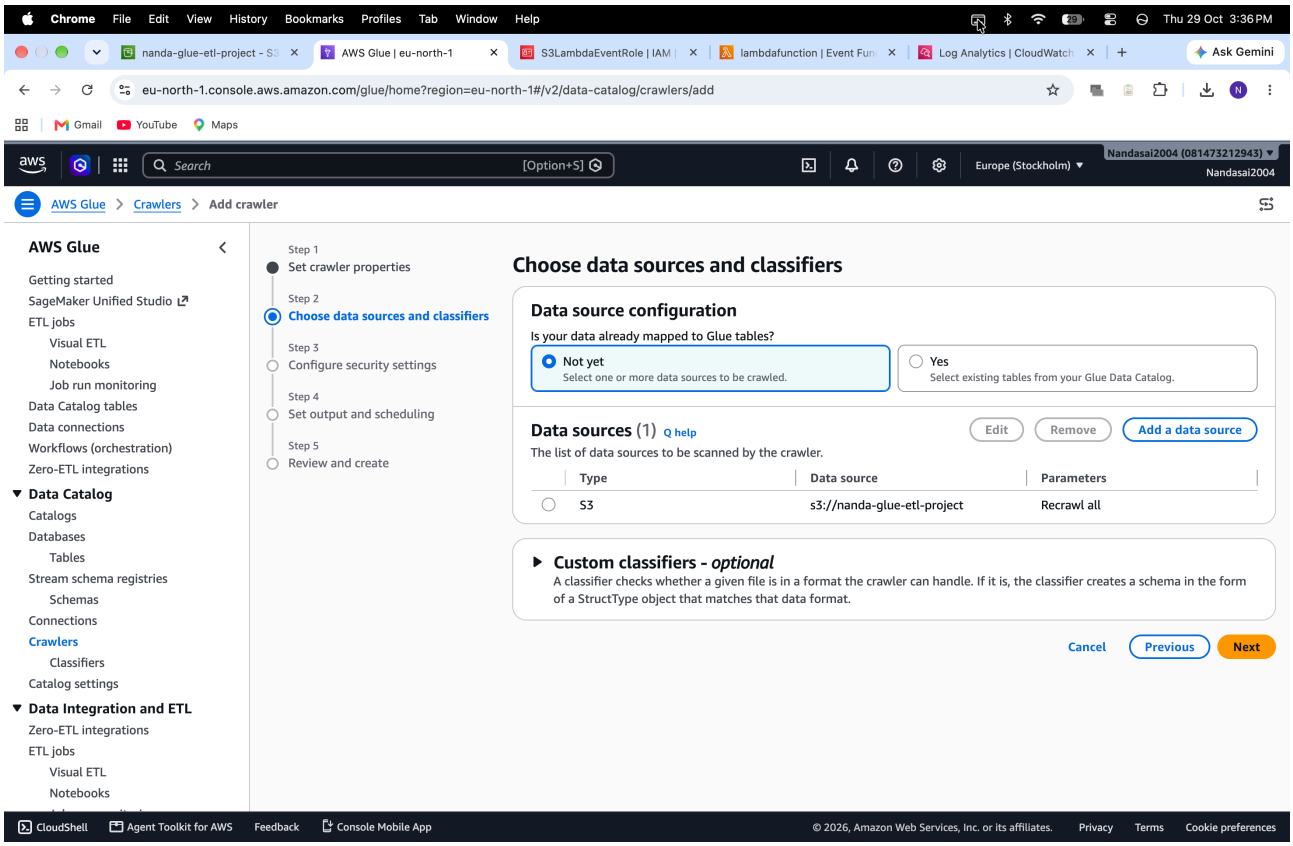
To design and implement an AWS Glue ETL Pipeline with Amazon S3 to extract employee data from CSV files, transform and clean the data using AWS Glue, and store the processed data in Amazon S3 for further analysis.

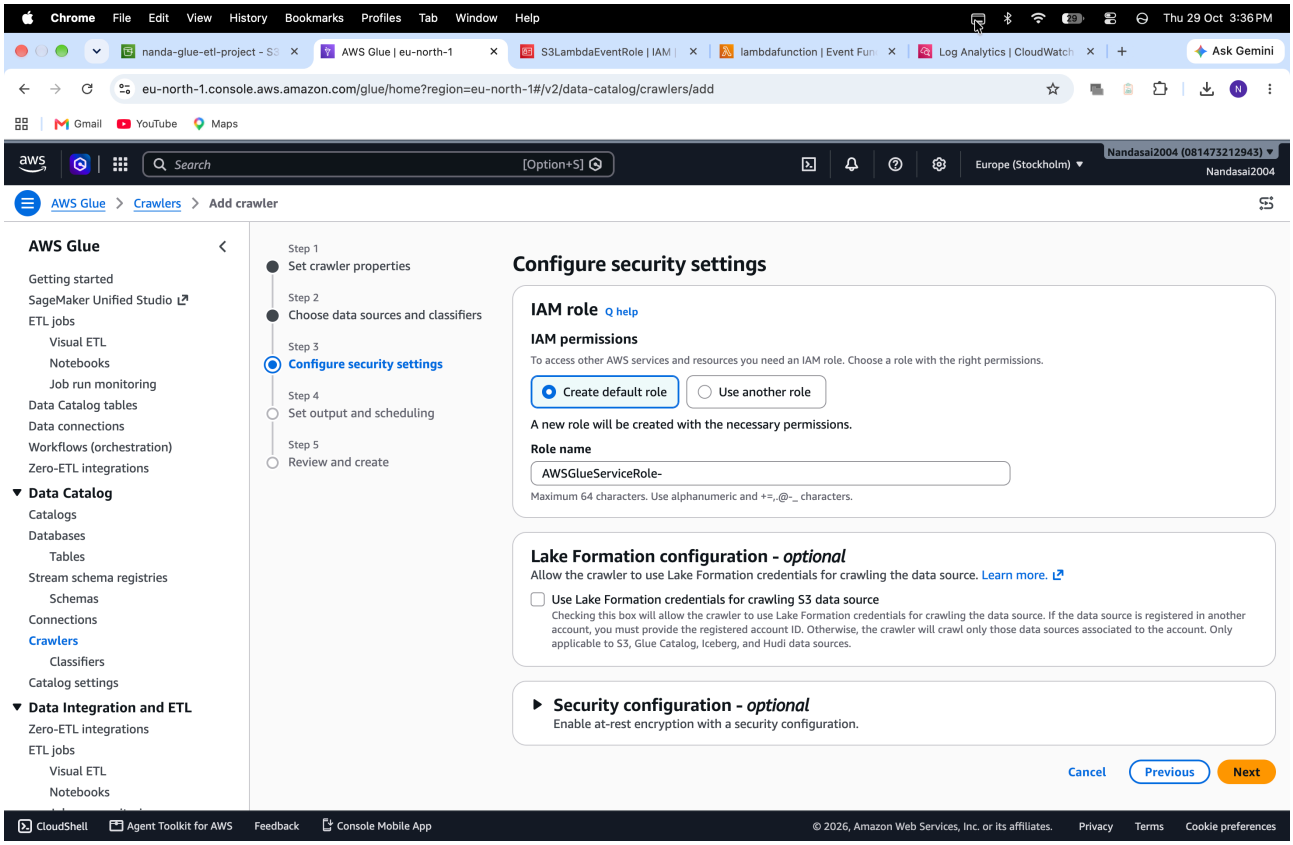
Step 1 — Prepare your S3 folders

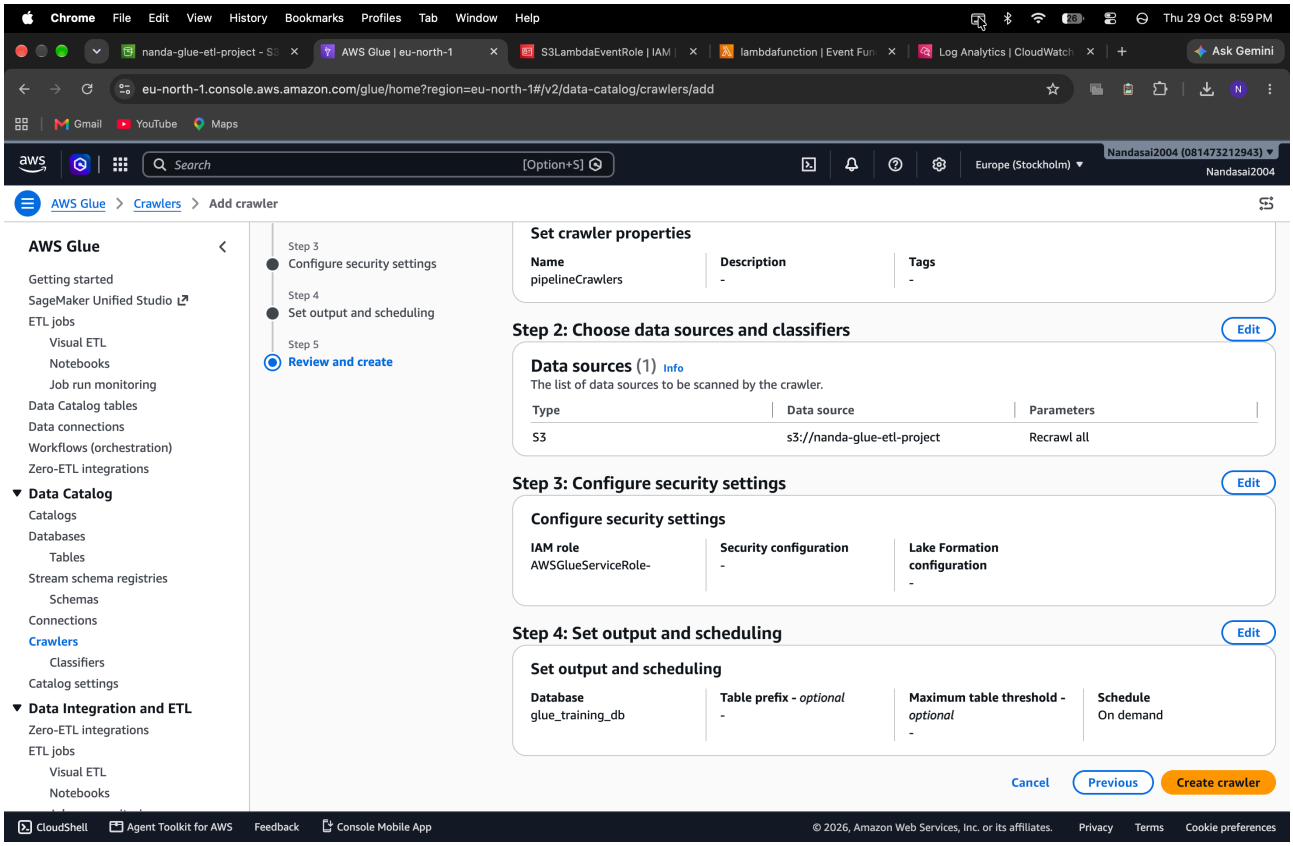




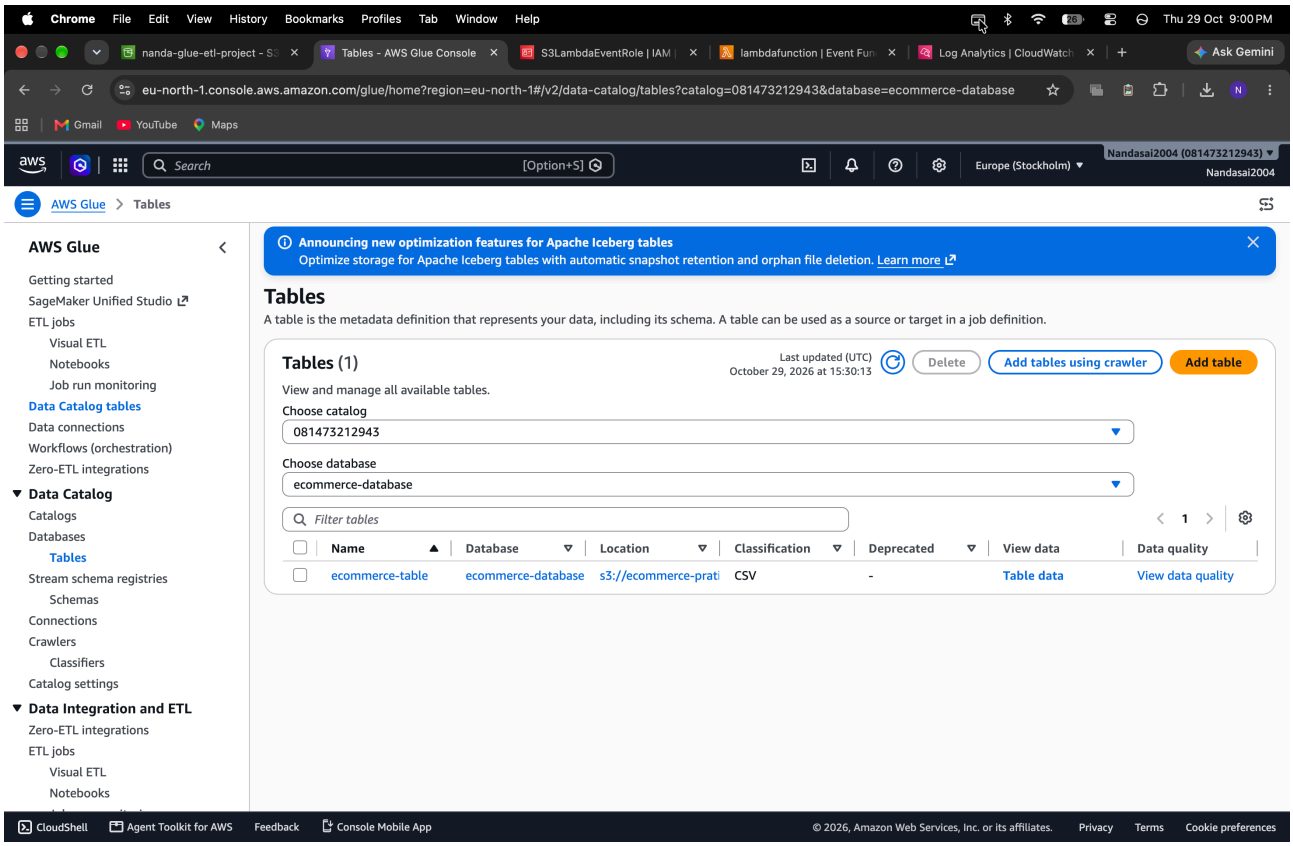
Step 2 — Create the Glue Data Catalog table



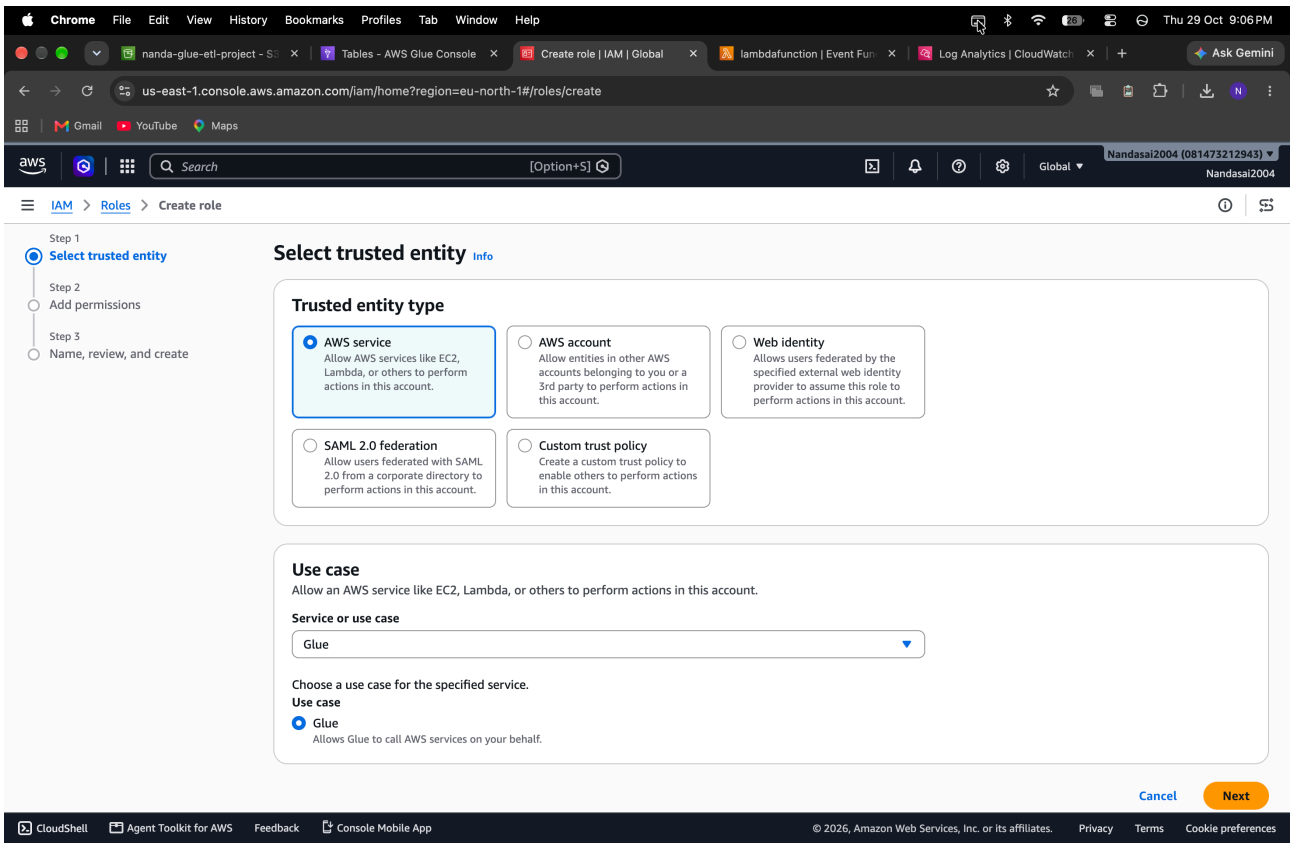

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 3:36 PM
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/crawlers/add
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Crawlers > Add crawler
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)
Zero-ETL integrations
Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings
Data Integration and ETL
Zero-ETL integrations
ETL jobs
Visual ETL
Notebooks
Step 1
Set crawler properties
Step 2
Choose data sources and classifiers
Step 3
Configure security settings
Step 4
Set output and scheduling
Step 5
Review and create
Configure security settings
IAM role Q help
IAM permissions
To access other AWS services and resources you need an IAM role. Choose a role with the right permissions.
Create default role
Use another role
A new role will be created with the necessary permissions.
Role name
AWSGlueServiceRole-
Maximum 64 characters. Use alphanumeric and +=,.@-_ characters.
Lake Formation configuration - optional
Allow the crawler to use Lake Formation credentials for crawling the data source. Learn more.
Use Lake Formation credentials for crawling S3 data source
Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.
Security configuration - optional
Enable at-rest encryption with a security configuration.
Cancel
Previous
Next
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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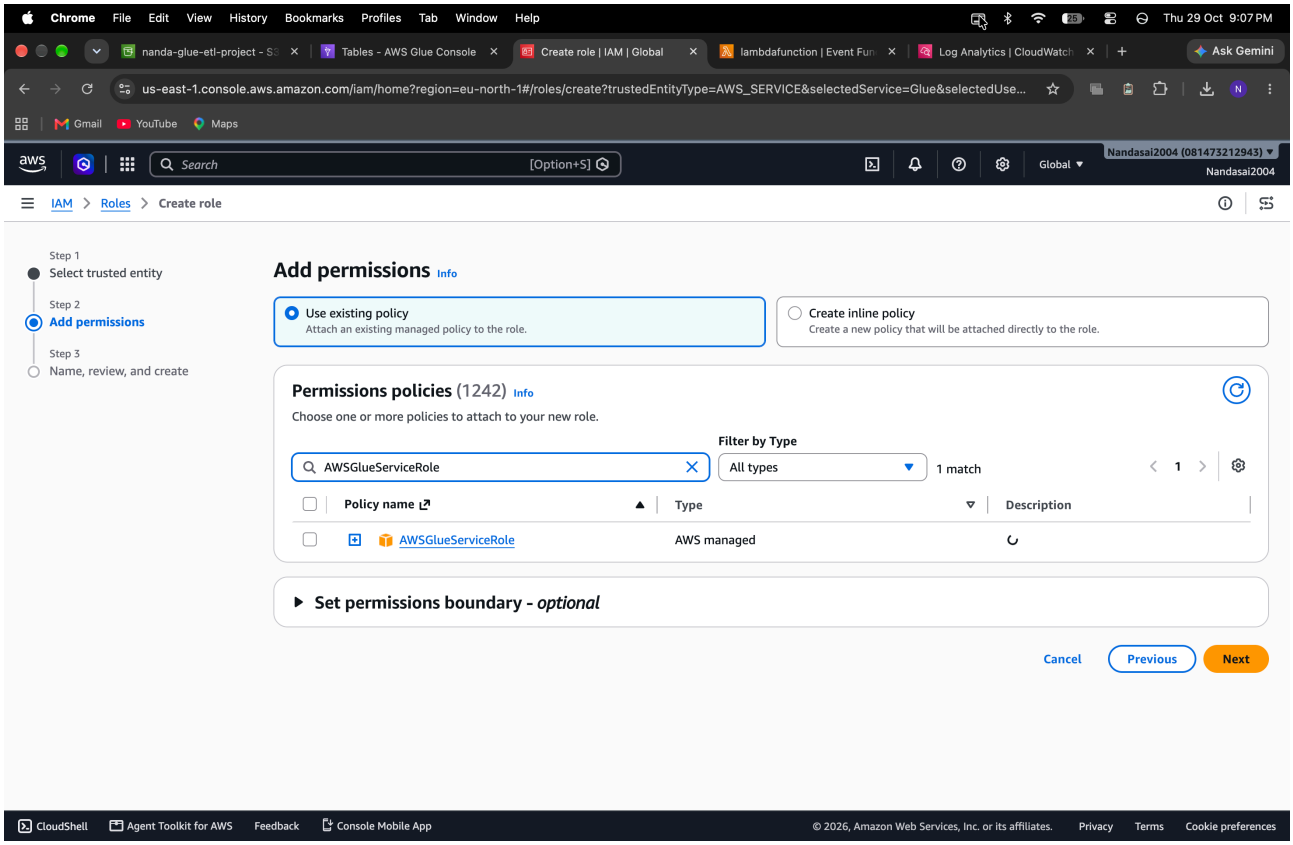

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 8:59 PM
eu-north-1.console.aws.amazon.com/glue/home?region=eu-north-1#/v2/data-catalog/crawlers/add
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
AWS Glue > Crawlers > Add crawler
AWS Glue
Getting started
SageMaker Unified Studio
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)
Zero-ETL integrations
Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings
Data Integration and ETL
Zero-ETL integrations
ETL jobs
Visual ETL
Notebooks
Step 3
Configure security settings
Step 4
Set output and scheduling
Step 5
Review and create
Set crawler properties
Name
pipelineCrawlers
Description
-
Tags
-
Step 2: Choose data sources and classifiers
Edit
Data sources (1) Info
The list of data sources to be scanned by the crawler.
Type
Data source
Parameters
S3
s3://nanda-glue-etl-project
Recrawl all
Step 3: Configure security settings
Edit
Configure security settings
IAM role
AWSGlueServiceRole-
Security configuration
-
Lake Formation configuration
-
Step 4: Set output and scheduling
Edit
Set output and scheduling
Database
glue_training_db
Table prefix - optional
-
Maximum table threshold - optional
-
Schedule
On demand
Cancel
Previous
Create crawler
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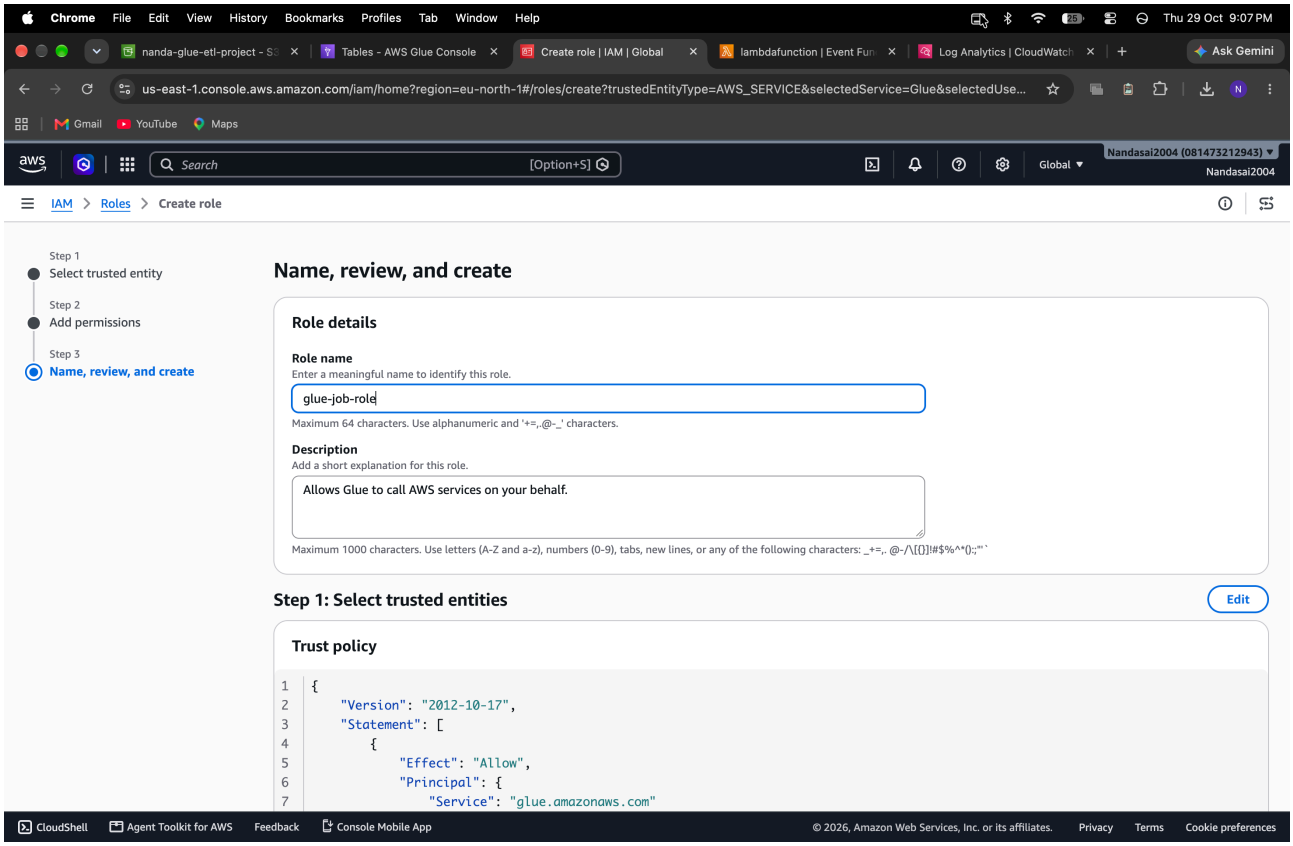
Step 3 — Configure IAM permissions.

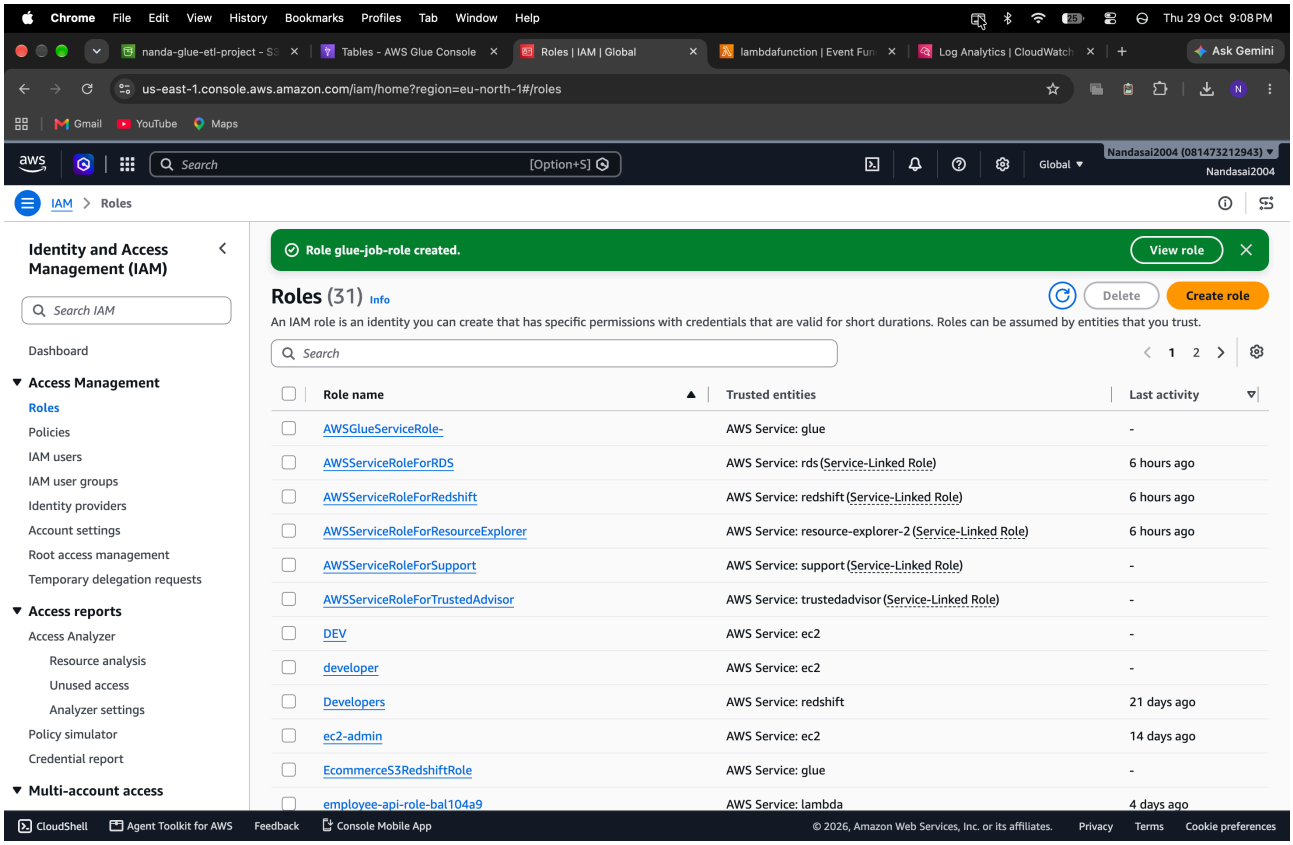


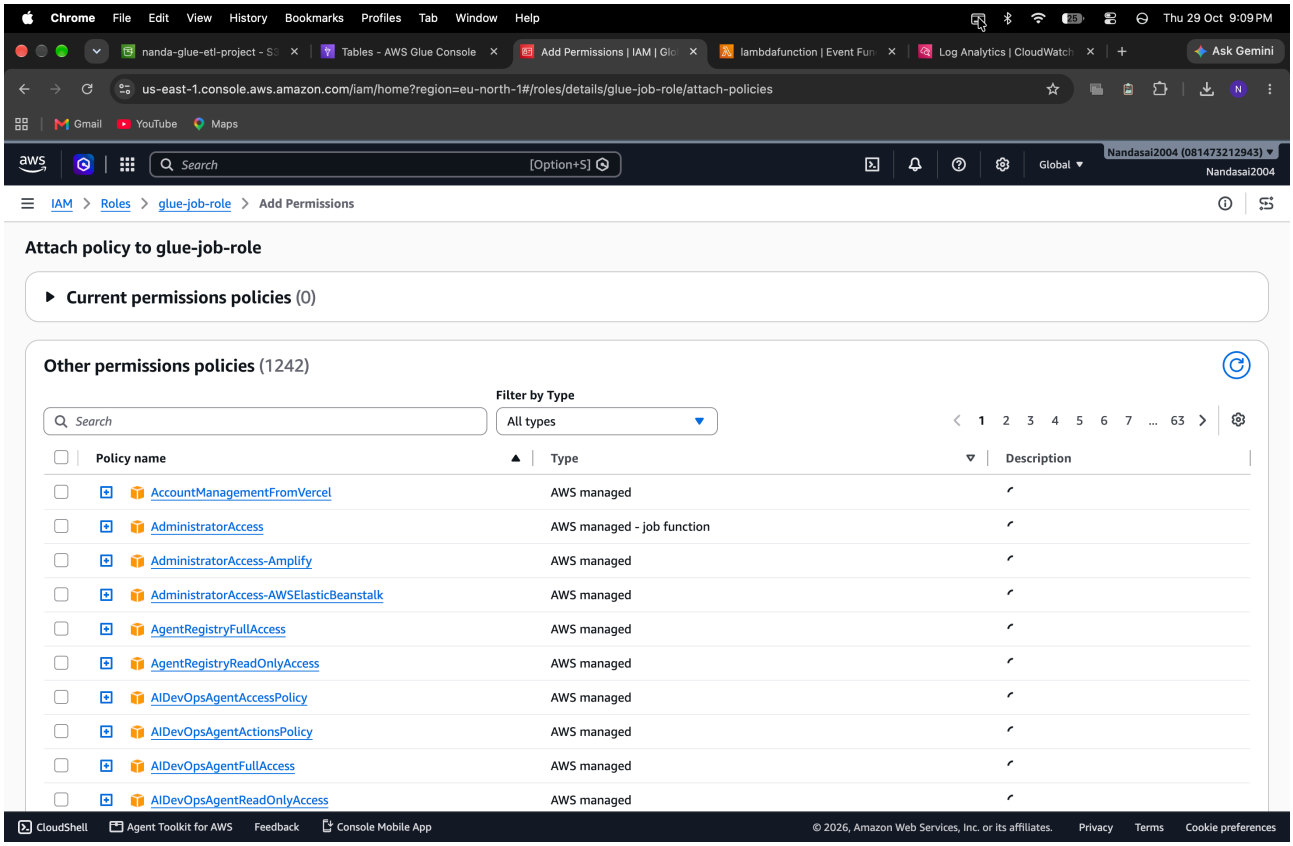
Step 3 — Configure IAM Permissions for AWS Glue

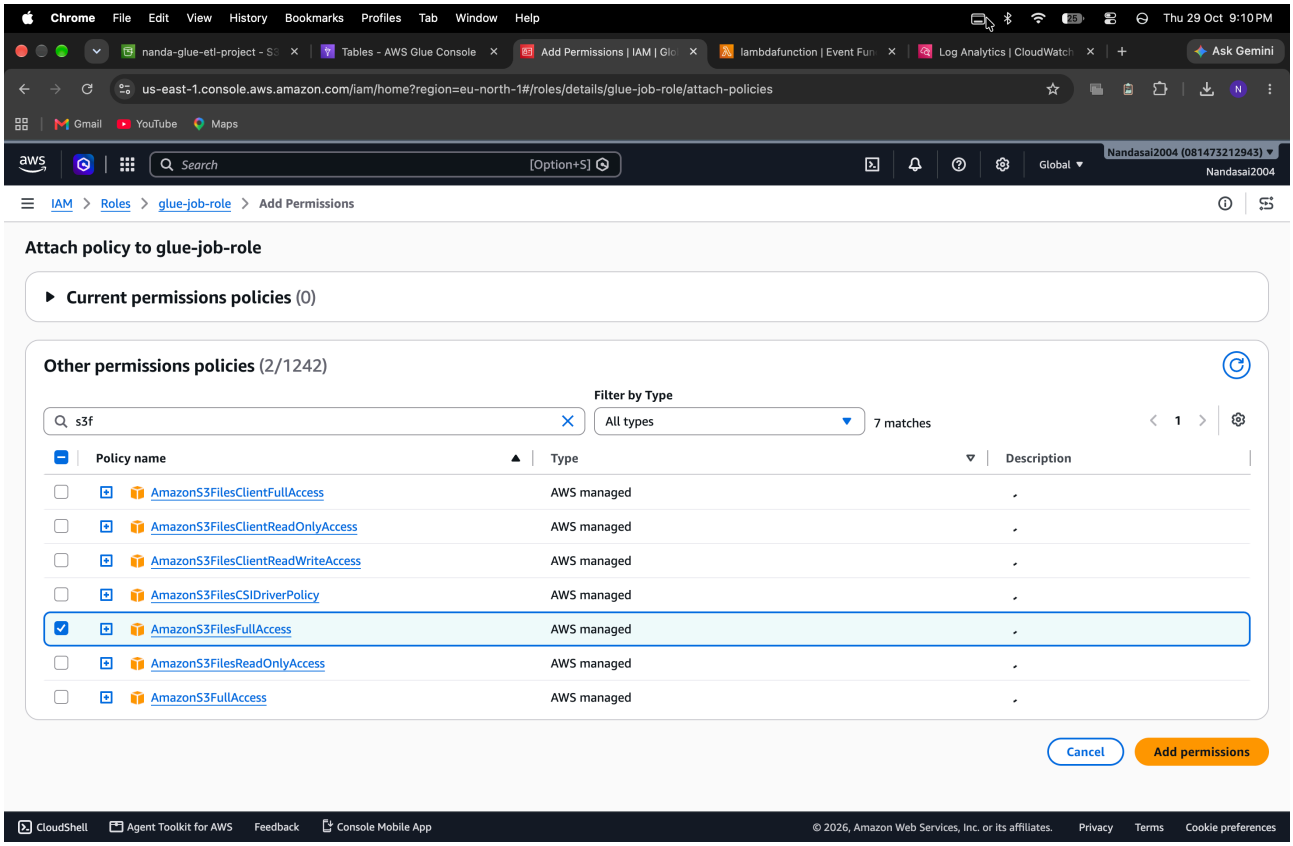


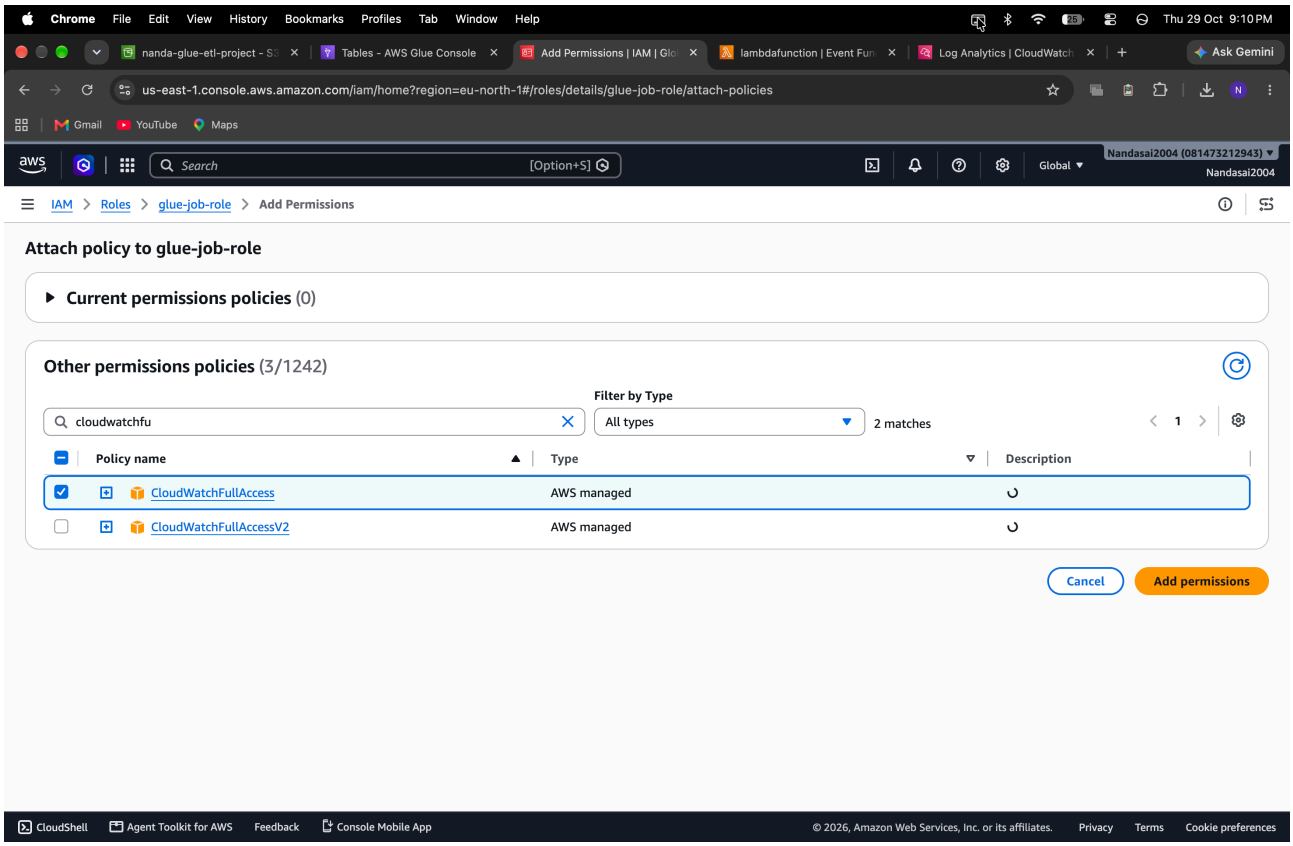

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:07 PM
nanda-glue-etl-project - S3
Tables - AWS Glue Console
Create role | IAM | Global
lambdafunction | Event Fun
Log Analytics | CloudWatch
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=Glue&selectedUse...
Gmail YouTube Maps
aws Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Use existing policy
Attach an existing managed policy to the role.
Create inline policy
Create a new policy that will be attached directly to the role.
Permissions policies (1242) Info
Choose one or more policies to attach to your new role.
Filter by Type
AWSGlueServiceRole
All types
1 match
1
Policy name
Type
Description
AWSGlueServiceRole
AWS managed
Set permissions boundary - optional
Cancel
Previous
Next
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Thu 29 Oct 9:07 PM
nanda-glue-etl-project - S3
Tables - AWS Glue Console
Create role | IAM | Global
lambdafunction | Event Fun
Log Analytics | CloudWatch
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=Glue&selectedUse...
Gmail YouTube Maps
aws Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Name, review, and create
Role details
Role name
Enter a meaningful name to identify this role.
glue-job-role
Maximum 64 characters. Use alphanumeric and '+=,.@-_' characters.
Description
Add a short explanation for this role.
Allows Glue to call AWS services on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,. @-/\[{}]!#$%^*();"'`
Step 1: Select trusted entities
Edit
Trust policy
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Principal": {
"Service": "glue.amazonaws.com"
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Agent Toolkit for AWS
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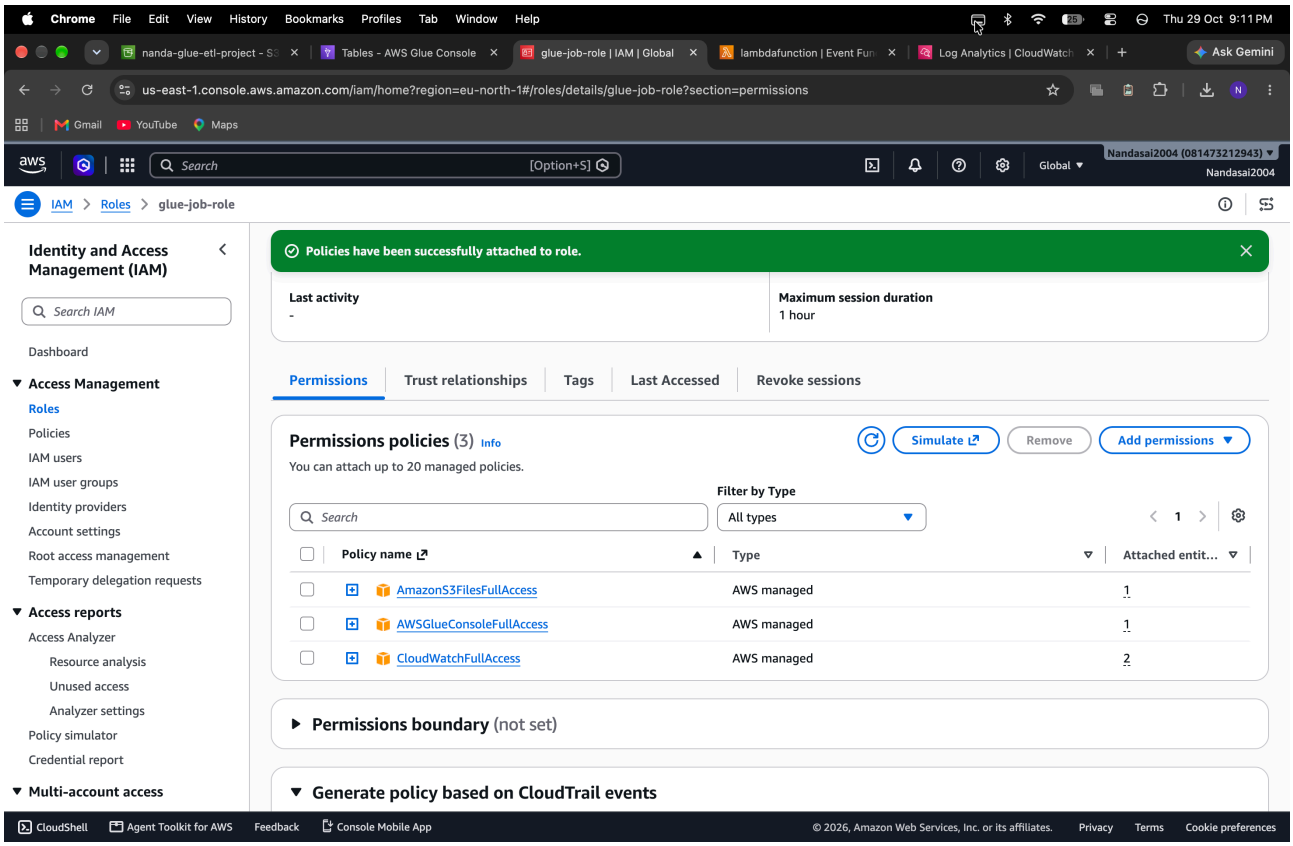

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:08 PM
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles
Nandasai2004 (081473212943)
IAM > Roles
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Role glue-job-role created.
View role
Roles (31) Info
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Delete
Create role
Role name
Trusted entities
Last activity
AWSGlueServiceRole-
AWS Service: glue
AWSServiceRoleForRDS
AWS Service: rds (Service-Linked Role)
6 hours ago
AWSServiceRoleForRedshift
AWS Service: redshift (Service-Linked Role)
6 hours ago
AWSServiceRoleForResourceExplorer
AWS Service: resource-explorer-2 (Service-Linked Role)
6 hours ago
AWSServiceRoleForSupport
AWS Service: support (Service-Linked Role)
AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Service-Linked Role)
DEV
AWS Service: ec2
developer
AWS Service: ec2
Developers
AWS Service: redshift
21 days ago
ec2-admin
AWS Service: ec2
14 days ago
EcommerceS3RedshiftRole
AWS Service: glue
employee-api-role-bal104a9
AWS Service: lambda
4 days ago
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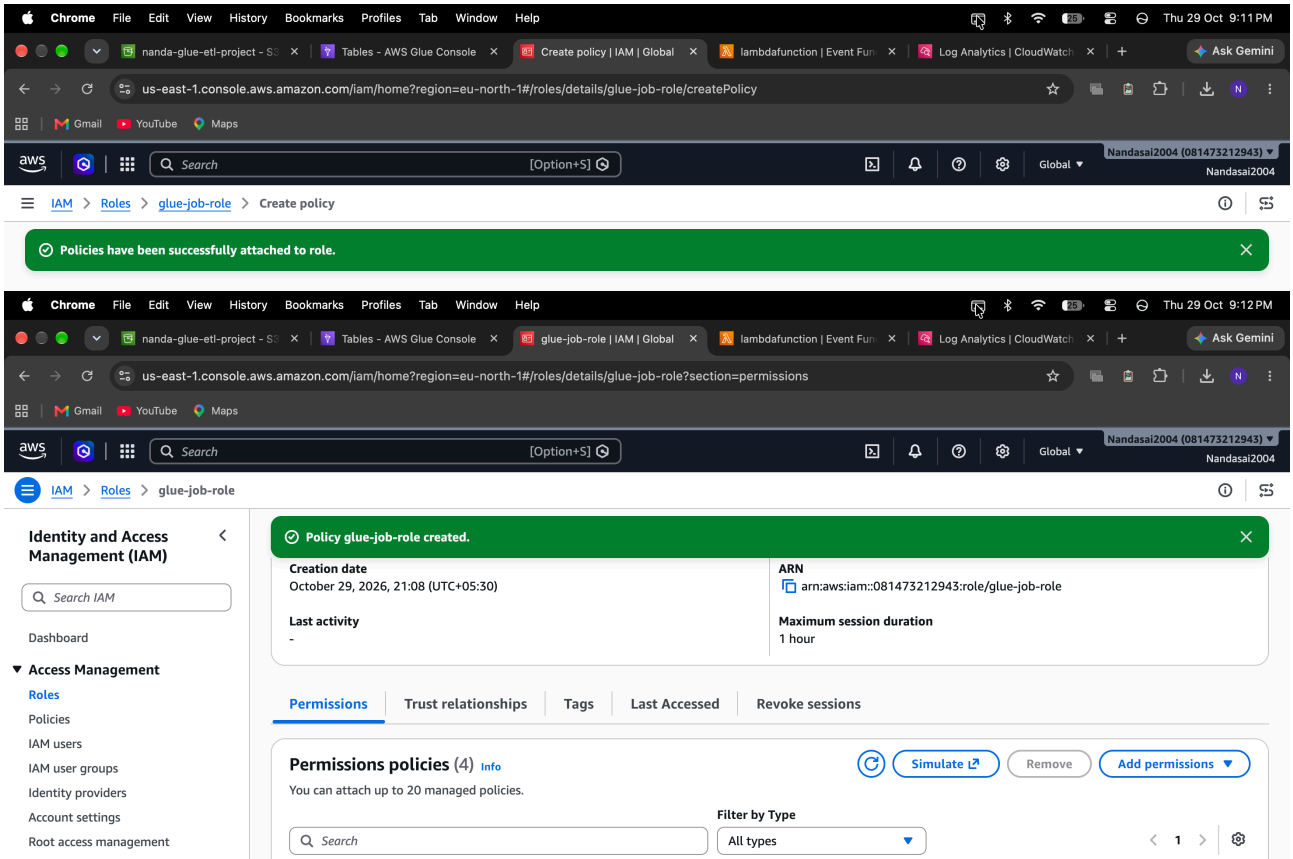

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:09 PM
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/details/glue-job-role/attach-policies
Nandasai2004 (081473212943)
IAM > Roles > glue-job-role > Add Permissions
Attach policy to glue-job-role
Current permissions policies (0)
Other permissions policies (1242)
Filter by Type
All types
Policy name
Type
Description
AccountManagementFromVercel
AWS managed
AdministratorAccess
AWS managed - job function
AdministratorAccess-Amplify
AWS managed
AdministratorAccess-AWSElasticBeanstalk
AWS managed
AgentRegistryFullAccess
AWS managed
AgentRegistryReadOnlyAccess
AWS managed
AIDevOpsAgentAccessPolicy
AWS managed
AIDevOpsAgentActionsPolicy
AWS managed
AIDevOpsAgentFullAccess
AWS managed
AIDevOpsAgentReadOnlyAccess
AWS managed
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Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:10 PM
nanda-glue-etl-project - S3
Tables - AWS Glue Console
Add Permissions | IAM | Glo
lambdafunction | Event Fun
Log Analytics | CloudWatch
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/details/glue-job-role/attach-policies
Gmail YouTube Maps
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > glue-job-role > Add Permissions
Attach policy to glue-job-role
Current permissions policies (0)
Other permissions policies (2/1242)
Filter by Type
s3f
All types
7 matches
1
Policy name
Type
Description
AmazonS3FilesClientFullAccess AWS managed
AmazonS3FilesClientReadOnlyAccess AWS managed
AmazonS3FilesClientReadWriteAccess AWS managed
AmazonS3FilesCSIDriverPolicy AWS managed
AmazonS3FilesFullAccess AWS managed
AmazonS3FilesReadOnlyAccess AWS managed
AmazonS3FullAccess AWS managed
Cancel
Add permissions
CloudShell
Agent Toolkit for AWS
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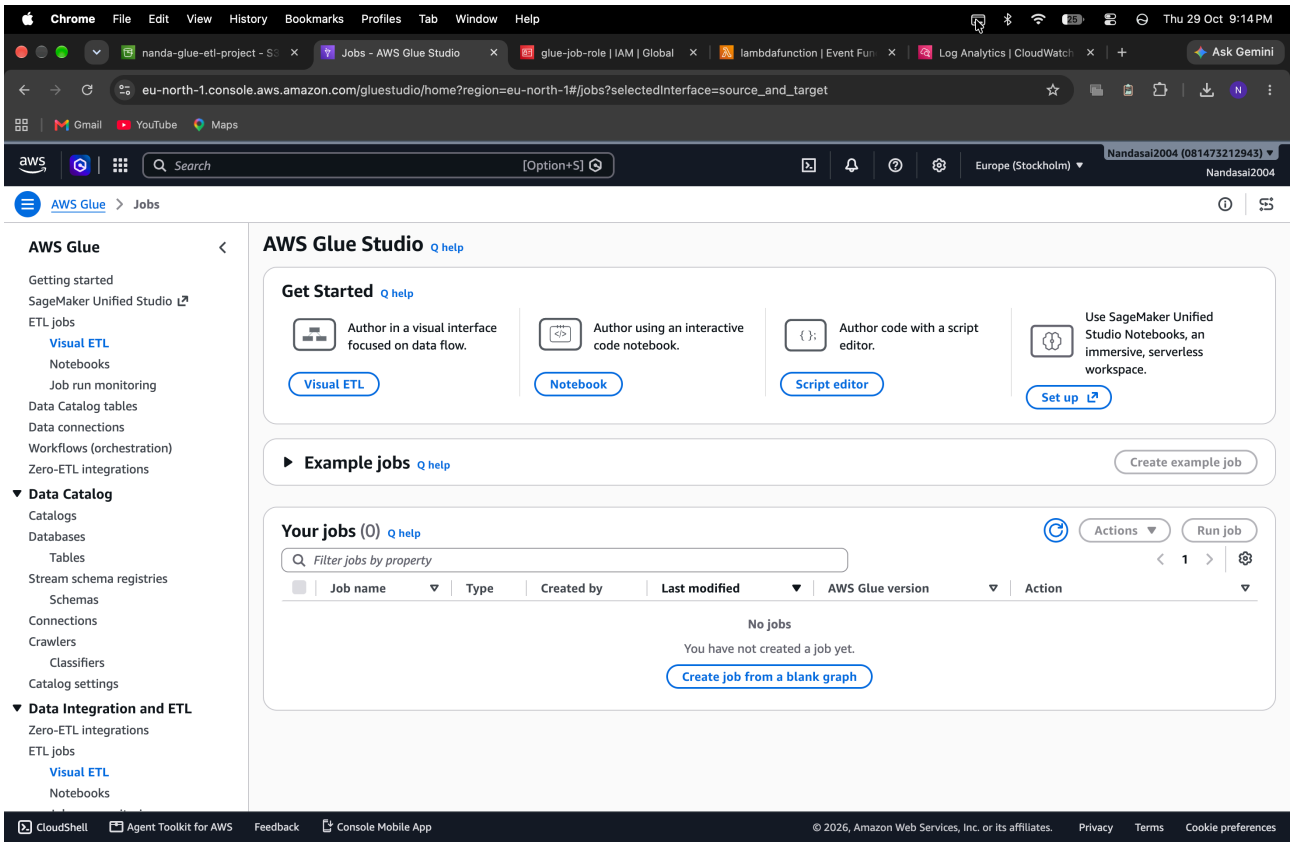

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:10 PM
nanda-glue-etl-project - S3
Tables - AWS Glue Console
Add Permissions | IAM | Glo
lambdafunction | Event Fun
Log Analytics | CloudWatch
Ask Gemini
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/details/glue-job-role/attach-policies
Gmail YouTube Maps
Search [Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Roles > glue-job-role > Add Permissions
Attach policy to glue-job-role
Current permissions policies (0)
Other permissions policies (3/1242)
Filter by Type
cloudwatchfu
All types
2 matches
1
Policy name
Type
Description
CloudWatchFullAccess AWS managed
CloudWatchFullAccessV2 AWS managed
Cancel
Add permissions
CloudShell
Agent Toolkit for AWS
Feedback
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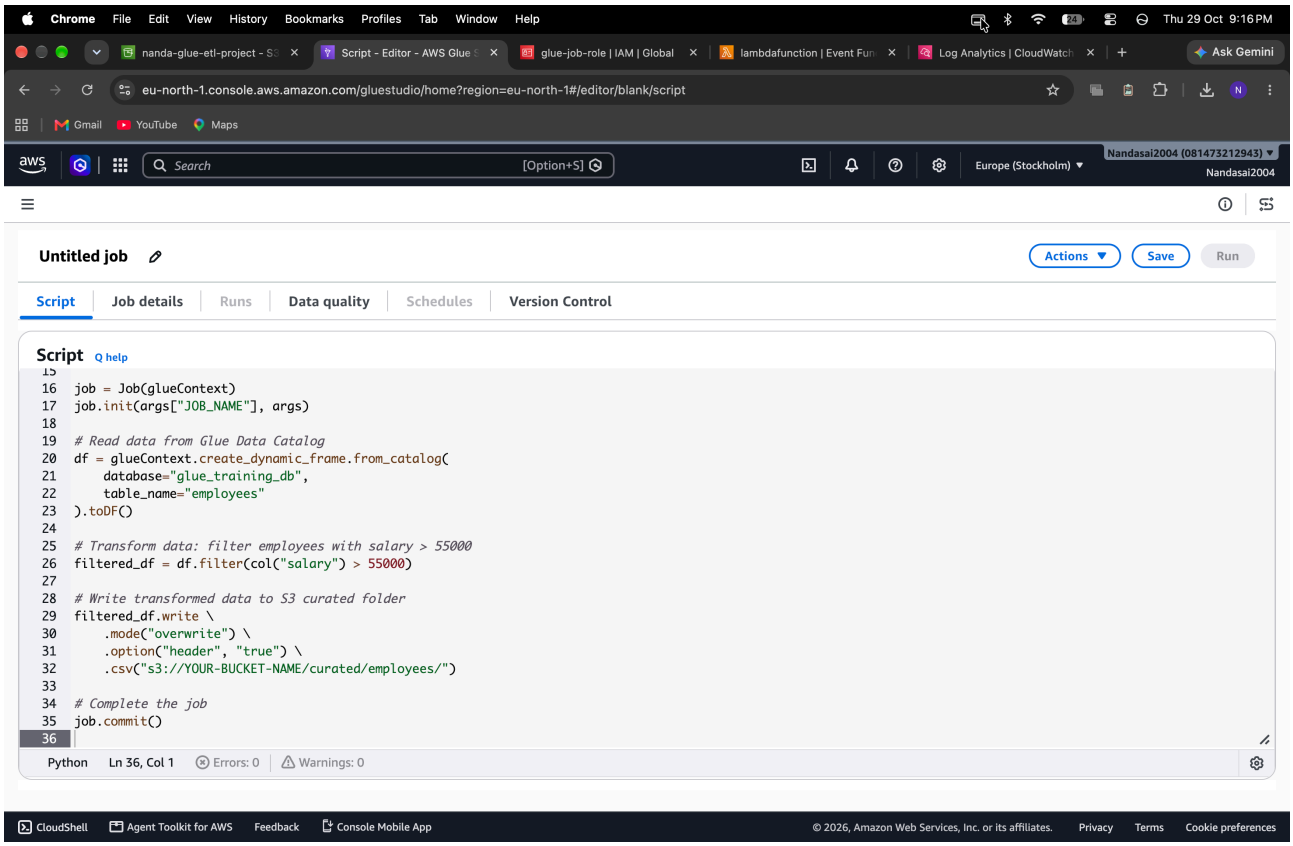
Step 4 — Create the ETL job.

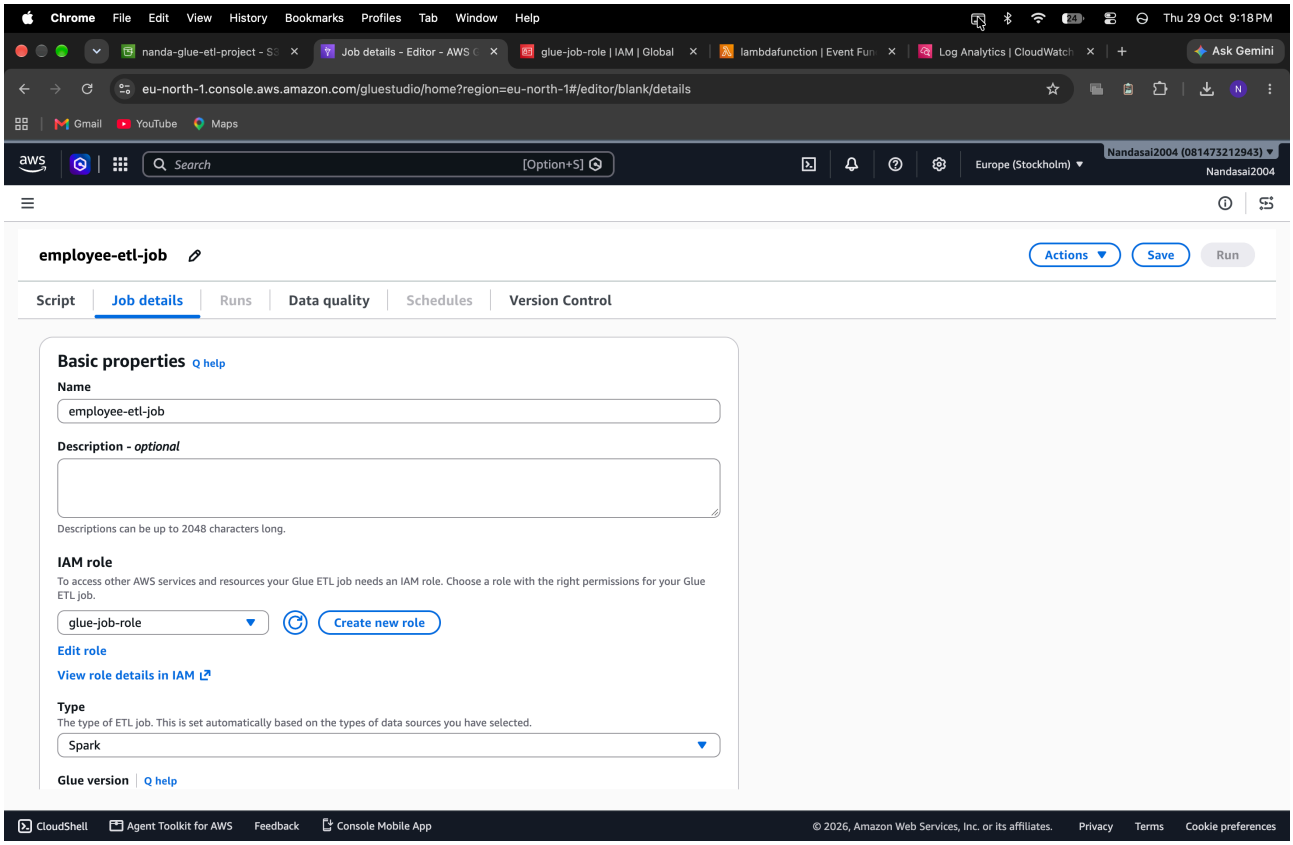


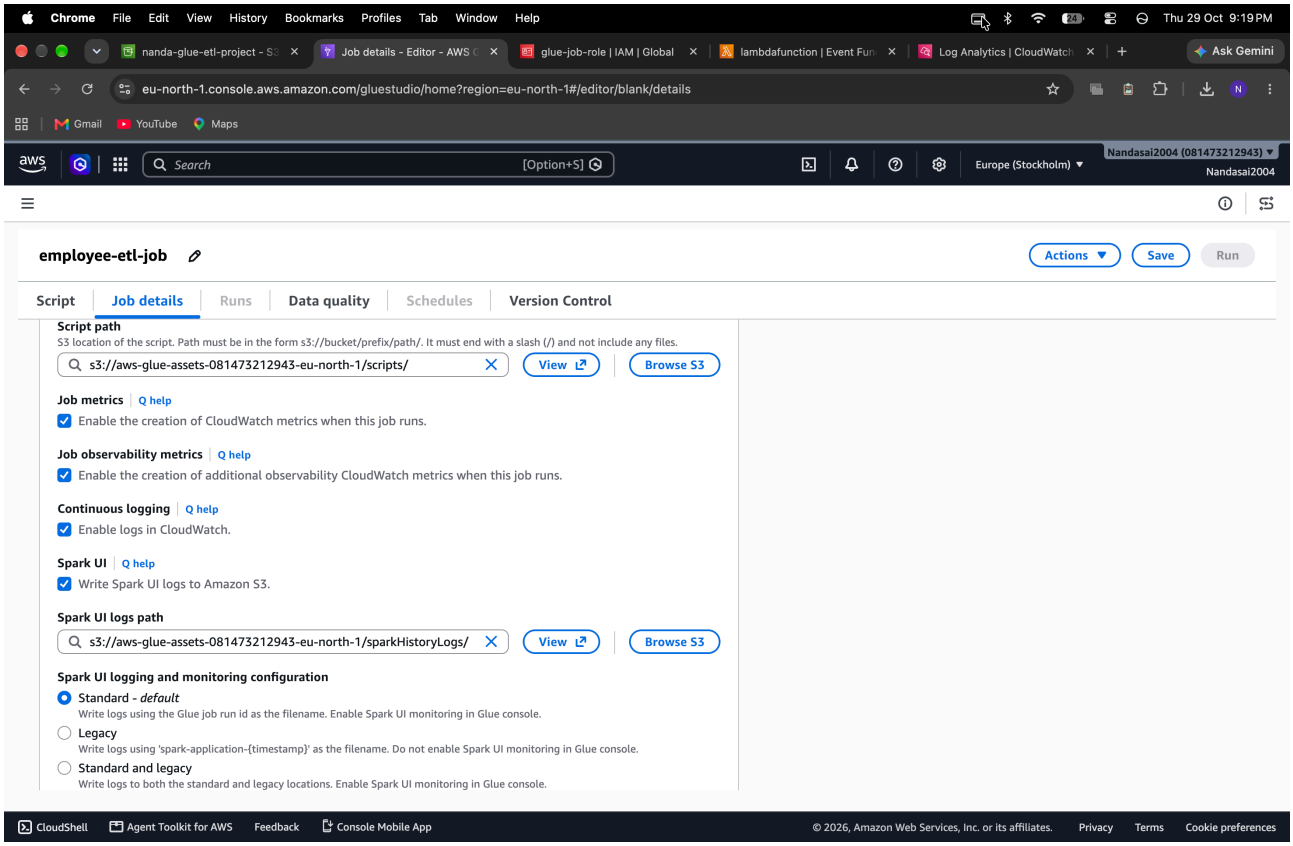


Step 4 — Create an AWS Glue ETL Job






Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:18 PM
nanda-glue-etl-project - S3
Job details - Editor - AWS
glue-job-role | IAM | Global
lambdafunction | Event Fun
Log Analytics | CloudWatch
Ask Gemini
eu-north-1.console.aws.amazon.com/gluestudio/home?region=eu-north-1#/editor/blank/details
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
employee-etl-job
Actions
Save
Run
Script
Job details
Runs
Data quality
Schedules
Version Control
Basic properties
Q help
Name
employee-etl-job
Description - optional
Descriptions can be up to 2048 characters long.
IAM role
To access other AWS services and resources your Glue ETL job needs an IAM role. Choose a role with the right permissions for your Glue ETL job.
glue-job-role
Create new role
Edit role
View role details in IAM
Type
The type of ETL job. This is set automatically based on the types of data sources you have selected.
Spark
Glue version
Q help
CloudShell
Agent Toolkit for AWS
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Chrome File Edit View History Bookmarks Profiles Tab Window Help
Thu 29 Oct 9:19 PM
nanda-glue-etl-project - S3
Job details - Editor - AWS
glue-job-role | IAM | Global
lambdafunction | Event Fun
Log Analytics | CloudWatch
Ask Gemini
eu-north-1.console.aws.amazon.com/gluestudio/home?region=eu-north-1#/editor/blank/details
Gmail YouTube Maps
Search [Option+S]
Europe (Stockholm)
Nandasai2004 (081473212943)
Nandasai2004
employee-etl-job
Actions
Save
Run
Script
Job details
Runs
Data quality
Schedules
Version Control
Script path
S3 location of the script. Path must be in the form s3://bucket/prefix/path/. It must end with a slash (/) and not include any files.
s3://aws-glue-assets-081473212943-eu-north-1/scripts/
View
Browse S3
Job metrics
Q help
Enable the creation of CloudWatch metrics when this job runs.
Job observability metrics
Q help
Enable the creation of additional observability CloudWatch metrics when this job runs.
Continuous logging
Q help
Enable logs in CloudWatch.
Spark UI
Q help
Write Spark UI logs to Amazon S3.
Spark UI logs path
s3://aws-glue-assets-081473212943-eu-north-1/sparkHistoryLogs/
View
Browse S3
Spark UI logging and monitoring configuration
Standard - default
Write logs using the Glue job run id as the filename. Enable Spark UI monitoring in Glue console.
Legacy
Write logs using 'spark-application-{timestamp}' as the filename. Do not enable Spark UI monitoring in Glue console.
Standard and legacy
Write logs to both the standard and legacy locations. Enable Spark UI monitoring in Glue console.
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